

Deliverable 3.1

Report on Requirement Analysis and Functional Specifications

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Definitions, Acronyms and Abbreviations

ACRONYMS / ABBREVIATIONS	DESCRIPTION
UI	User Interface
UX	User Experience
GUI	Graphical User Interface
FAQ	Frequently Asked Questions
WCAG	Web Content Accessibility Guidelines
LLM	Large Language Model
GDPR	General Data Protection Regulation

Modal Verbs Nomenclature [1][2]

1. **MUST** This word, or the terms "REQUIRED" or "SHALL," means that the definition is an absolute requirement of the specification.
2. This phrase "SHALL NOT," means that the definition is an absolute prohibition of the specification.
3. **SHOULD / SHOULD REQUIRED** This word, or the adjective "RECOMMENDED," means that in particular circumstances, there may be valid reasons for ignoring a particular element, but that it is necessary to understand all of its implications and consider them carefully before choosing a different path.
4. The phrase **SHOULD NOT**, or the phrase "NOT RECOMMENDED," means that in particular circumstances, there may be valid reasons why the behaviour is acceptable or even useful, but that you need to understand all the implications and carefully evaluate the case before implementing any behaviour described with this label.
5. **MAY** This word, or the adjective "OPTIONAL," means that an element is genuinely optional. One vendor may include the aspect because a particular market requires it or because it believes it improves the product, while another vendor may omit the same element.

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1. Background of the CREA3 project

Predictive justice and digitalisation are the main topics of CREA3 Project, which can be used in family disputes in national and European contexts. CREA3 aims to introduce AI-driven tools to assist EU citizens in resolving disputes by applying innovative game theoretical (GT) algorithms. To improve access to justice and interaction with the rule of law, we seek to develop a strategy that addresses harmonisation across digitalisation and predictive algorithms. This approach, which combines legal research with extensive text-mining techniques, is novel since it will explore the idea of testing and evaluating how well family law systems in 6 EU legislations adhere to the new forthcoming “Regulation on the digitalisation of judicial cooperation and access to justice in cross-border civil, commercial and criminal matters”. In particular, the CREA3 project will focus on the following needs: 1. Working with e-communication means and documents with e-signatures from the target groups involved (individuals, legal professionals and courts). 2. Allowing parties and other relevant actors in family hearings to participate through videoconferencing. 3. Guaranteeing access to justice for vulnerable people who need technical assistance or may not have the required skills. This ensures access to justice for citizens whose personal capacity does not allow them seamless access to digital tools. Furthermore, CREA3 represents the final and natural conclusion of the already developed CREA (G.A. 766463) and CREA2 (G.A. n. 101046629) projects, aimed to specifically improve the digitalisation and access to justice among 6 different Member States in family law matters. The indicated target groups will have access to this novel approach's functionalities through a user-friendly web interface. The CREA3 Consortium will also focus on the interoperability of the system in the six legislations involved, exchange and spread of knowledge about the new interface, the chatbot and the algorithms.

2. Scope and objectives of the document

The CREA3 Project aims to modernize civil law procedures throughout the European Union, enabling citizens, legal professionals, judicial authorities, and, in particular, vulnerable users to achieve swifter, fairer, and non-discriminatory access to justice. Its strategic vision aims to augment the performances of its predecessors, CRE and CREA2, seamlessly introducing electronic signatures, authenticated videoconferences with automated evidentiary logging, and third, providing inclusive access measures that support individuals with limited digital literacy and cognitive capabilities.

Specifically, this document establishes the framework that will guide the design and implementation of the CREA3 Smart Platform, embracing every facet of the required digital service.

It starts by reviewing the legal and procedural landscape that the platform must support, identifying the users who will interact with it, and distilling the technological enablers required to ensure compliance and usability. Its purpose is to translate the project's overarching ambition into a coherent set of high-level requirements that all subsequent technical and organizational work must observe.

In detail, this report pursues five complementary objectives:

- It defines the UI/UX principles underpinning the graphical interface and the integrated conversational agent.
- It defines the platform's stakeholders and user profiles, focusing on special groups with cognitive, visual, or skill limitations.
- It enumerates some of the functional capabilities that the platform must expose, ranging from user registration and authentication through the staged life-cycle of the digital dispute.
- It captures the conversational chatbot's non-functional expectations or knowledge requirements, such as multilingual understanding and user nationality identification.
- It signals areas and case studies—most notably the conversational agent's knowledge architecture—with a detailed workflow schema.

While the analysis is sufficiently comprehensive to guide architects and developers, this document intentionally avoids design details that would constrain innovation and later-stage developments, offering a practical reference for the expected outcomes for consortium partners, external consultants, and oversight bodies.

Briefly, report presents the baseline against which implementation milestones and acceptance criteria must be adhered to, ensuring that the emerging platform remains aligned with legal mandates, technological best practices, and the project's commitment to inclusive, EU-wide access to justice.

3 User Categories & Accessibility Principles

This section outlines the general user persona, characterized by a set of skills, roles, and possible impairments. It addresses their specific challenges and needs through a homogeneous set of design principles, following the Modal Verbs Nomenclature [1, 2], and lays the groundwork for the following sections and future developments.

3.1 Users Categories

The CREA3 platform SHOULD be specifically designed to support large-scale usage and accommodate diverse users. Three primary user categories within this ecosystem are considered, each with distinct roles and needs.

1. **Private citizens:**
Subjects acting in their own right, involved or not in a real dispute resolution, demanding assistance through the digitalized procedure.

Key characteristics:

- No formal legal training
- May be claimants, respondents, or general information seekers
- Seek guidance, clarity, and access to simplified legal processes

2. **Practitioners & Legal Experts:**
Legally literate individuals using the platform in a professional or administrative capacity. They may or may not be directly involved in a resolution process.

Subcategories:

- Mediators/Facilitators: Neutral third parties managing or guiding resolution between other parties
- Advisors/Legal Counsel: Professionals offering legal guidance without formal representation roles

Key characteristics:

- Possess legal knowledge or training
- Familiar with procedural frameworks

3. **Vulnerable Users**
Individuals, including private citizens and practitioners, facing barriers that affect their ability to use the platform effectively.

Possible vulnerabilities include:

- Cognitive (e.g., attention, memory, learning difficulties, dyslexia)
- Sensory (e.g., visual or hearing impairments)
- Linguistic (e.g., low literacy or language mismatches)
- Technological (e.g., limited access to digital tools or low digital literacy)

3.2 Design Principles

The design SHOULD be user-centric and fully aligned with the CREA3 project's principles of equity, accessibility, and justice. The system will be deployed as a web application that is navigable through structured menus and user interfaces.

All user actions will be initiated via the graphical user interface (GUI), and backend systems will enforce validation, access rights, and workflow rules.

- | | | | |
|----|----------------------|------------|--------------------|
| a) | Accessibility | and | Inclusivity |
|----|----------------------|------------|--------------------|
- The platform **MUST** conform to international accessibility standards (e.g., WCAG 2.1 Level AA) to guarantee usability for all users, including those with cognitive, visual, auditory, or motor impairments.
 - All text and graphical elements **SHOULD** use high-contrast color schemes to ensure readability and **MUST** be scalable to support screen magnification and assistive reading technologies.
 - The interface layout, font sizes, and touch target dimensions **MUST** be optimized for accessibility across all user categories, including those using mobile devices or assistive hardware.
 - All interactive components—including menus, buttons, links, and forms—**MAY** support keyboard navigation and be fully compatible with screen readers.
- | | | | | |
|----|------------------|------------|---------------|--------------------|
| b) | Usability | and | Visual | Consistency |
|----|------------------|------------|---------------|--------------------|
- User flows **SHOULD** be intuitive, with clearly labeled buttons, consistent navigation structures, and visual cues to guide progression.
 - Font sizes, touch targets, and layout spacing **MUST** be optimized to guarantee experiences of all user categories.
 - Icons and interface elements **SHALL NOT** rely solely on color to convey meaning, text, hidden text, labels, symbols, or patterns **MUST** be used to supplement color-based indicators.
 - Users **MUST** be provided with immediate and understandable feedback for every interaction (e.g., confirmations, warnings, errors).
- c) Multilingual and Multicultural Support**
- The platform **MAY** supports multiple languages relevant to the CREA3 participating countries, and **MAY**, accordingly, allows users to switch language settings without losing session progress.
 - All displayed text, labels, and messages **MUST** be understandable for all user categories[[Sec. 3.1](#)], leveraging the formality of legal terminologies without any impact on the UX of the users without formal legal training.
- d) Customization and Adaptability**
- The interface **MAY** allow user-specific settings (e.g., font size, contrast mode) to be configured and stored across sessions.
- e) Procedural Guidance and Support**
- Users **MUST** be supported by contextual help, tooltips, and inline guidance during complex tasks or legal form submissions.
 - A help system or chatbot assistant **SHOULD** be available to answer procedural questions, provide navigation help, and escalate to human support when necessary.

3. Functional Requirements

The platform **MUST** admit users through an electronic registration process that captures identity, role, and jurisdictional context and **SHOULD** thereafter enforce secure, role-based authentication at each login. Once authenticated, a user can create a dispute record that encapsulates the parties, the nature of the claim, and the governing procedural framework. The platform shall permit the ongoing management of each dispute, including the submission of evidence, the scheduling of hearings, and the monitoring of deadlines

while allowing real-time association of authorized agents as parties' representatives. Assets and goods relevant to enforcement or settlement SHOULD be recordable and maintainable within the same dispute context.

The disputes pass through a defined internal sequence.

- The initialization phase records the procedural basis and validates jurisdiction.
- The second phase captures agent preferences, encompassing language, preferred means of service, and accommodation requirements.
- The third phase evaluates the solution by correlating the dispute profile with available court routes and identifying any mandatory pre-action protocols.
- The fourth and final phase registers the agreement or adjudication outcome, applies qualified signatures, and stores the result immutably (possibly integrating in a chain system).

A context-sensitive conversational agent based on large language models (LLMs) remains a persistent overlay throughout the digital journey. It interprets free-text or possibly voice prompts, displays the corresponding procedural step on screen, and provides explanatory references to the underlying legislation.

3.1 Platform-Centric Specifications

This section is devoted to providing a set of requirements and platform specifications that describe how the CREA3 platform SHOULD behave and appear from a user's perspective. Generally speaking, the system will be implemented as a web application navigable through menus and interfaces. Users can initiate all functionalities through the GUI, while the backend logic will enforce validation, access, and consistency. The main elements of the platforms are presented in the Fig. 1 and briefly described in the Tab.1:

Section	Title	Description
Sec. 3.1.1	Landing Page	Entry point with general info and access options.
Sec. 3.1.2	User Registration Page	Allows new users to create an account.
Sec. 3.1.3	User Login Page	Authenticates returning users to access the system.
Sec. 3.1.4	Personal Home Page	User dashboard with navigation to main features.
Sec. 3.1.5	Create New Dispute	Start and define a new legal dispute.
Sec. 3.1.6	Managing Existing Disputes	View, edit, or delete active disputes.
Sec. 3.1.7	Agent Management	Add or modify participants in a dispute.
Sec. 3.1.8	Goods & Assets Management	Enter and manage disputed items or assets.
Sec. 3.1.9	Validation Phase	Final review before confirmation and mediation
Sec. 3.1.10	Video Conference Integration	Launch or join virtual dispute resolution sessions.

Section	Title	Description
<u>Sec. 3.1.1</u>	Landing Page	Entry point with general info and access options.
<u>Sec. 3.1.2</u>	User Registration Page	Allows new users to create an account.
<u>Sec. 3.1.11</u>	FAQs & Help	Access help topics and platform guidance.

Tab.1 CREA3 Platform's Elements Overview & Description

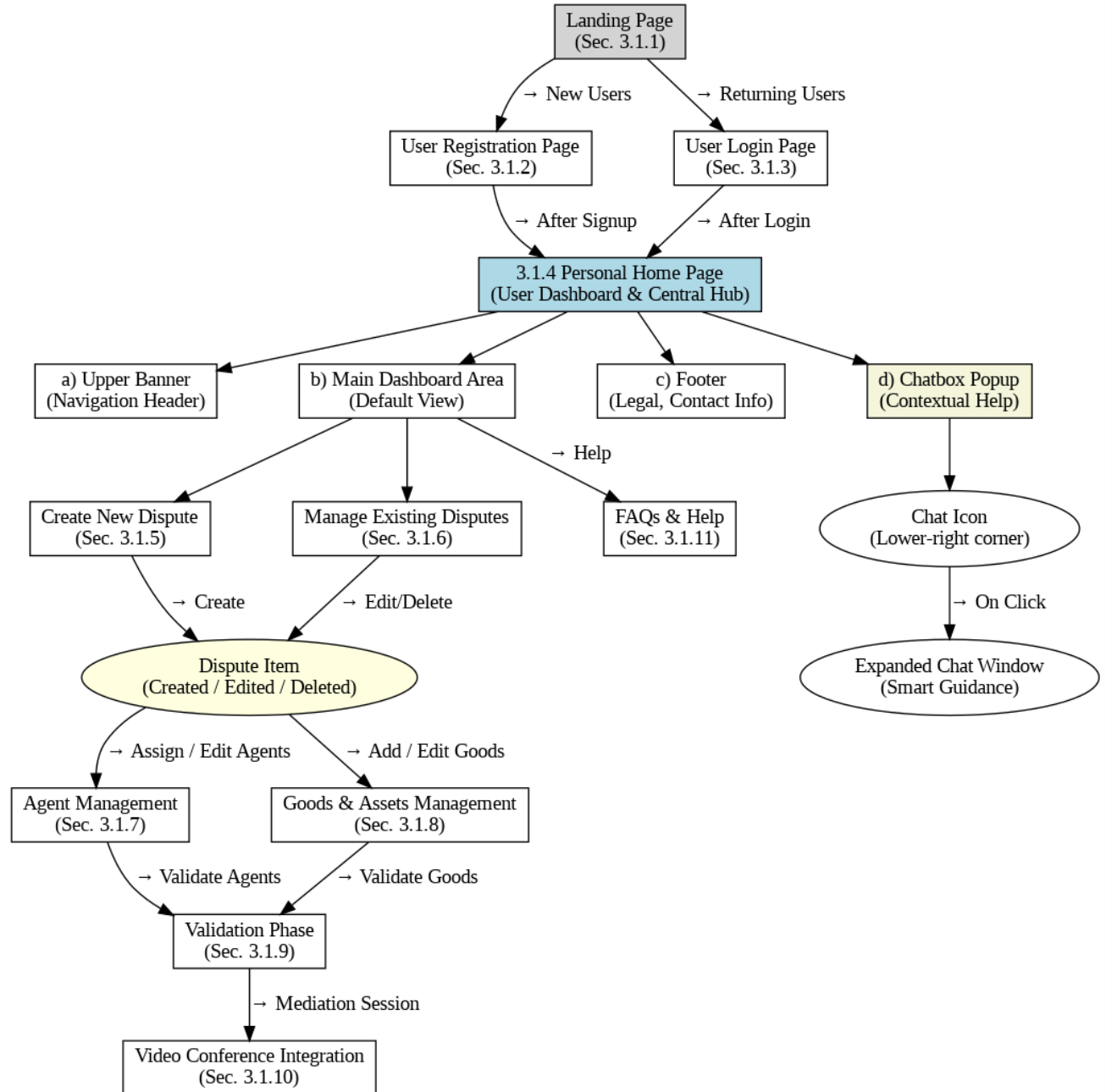


Fig. 1 Schematic Representation of the CREA3 Platform Modules

3.1.1 Landing Page

The Landing Page is the primary entry point to the CREA3 platform. It provides users with immediate orientation regarding the purpose of the platform, and access to key areas of content.

a) Functional	Description:	Structure	and	Layout
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The landing page is structured into distinct, clearly navigable areas, including:

- A top navigation bar offering access to sections such as:
 - Project overview
 - Research outputs
 - News and events
 - Contact page
- A prominent **call-to-action button** that directs users to the CREA3 platform solution area (e.g., login or dashboard)
- An **introductory section** (hero area) briefly outlining the mission of the CREA3 platform
- A **footer** providing legal information, funding acknowledgments, and external links

b) Visual	Identity
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The landing page will reflect the CREA3 branding, incorporating:

- A visually harmonious color palette
- A clean, minimalistic layout that emphasizes content clarity
- Appropriately placed logos and project branding elements

c) Content	Management
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The textual and visual content of the landing page should be **easily updatable** (preferably via a backend CMS or structured configuration file) to accommodate changes in project information, partners, or events.

3.1.2 User Registration Page

Allow new users to create an account on the CREA3 platform.

a) User	Interface
----------------	------------------

- Accessed via: **Navigation bar → "Register"**
- Displayed Fields:
 - First Name (*text input*)
 - Last Name (*text input*)
 - Email (*validated, unique*)
 - Password + Confirm Password (*minimum complexity enforced*)
 - Terms & Conditions acceptance checkbox (*view terms option with hyperlink*)

b) Frontend

Behavior

- Client-side validation on every field.
- Submit button is disabled until all fields are correctly filled.

c) Backend

Behavior

- Checks if user already exists by email or ID.
- If logged in, redirects to profile page with notice:
"You're already logged in as Mario Rossi. [[Log out](#)] to create a new account."

d) Postcondition

Upon successful submission, the user is redirected to a **Login page** Sec 3.1.3

3.1.3 User Login Page

Allow existing users to authenticate and access private sections and dispute panels.

a) User Interface

- Accessed via: **Go to CREA3 Solution → "Login form"**
- Fields:
 - Email/Username
 - Password
 - Remember Session Checkbox
- Options:
 - "Forgot Password?" (triggers email recovery flow)
 - "Create Account" (redirects to registration)

b) Backend Processing

- Server-side check against stored credentials
- Session token issued on success.

c) Postcondition

- Successful login redirects to the user dashboard
- Failed login shows an error with a retry prompt.

3.1.4 Personal Home Page

After logging in, users are redirected to their personal dashboard, which serves as the operational hub for managing disputes. It emphasizes usability, personalization, and accessibility.

a) *Upper Banner (Navigation Header)*

A persistent fixed and non-scrollable top navigation bar that spans the full width of the screen.

Key Elements:

- Platform Logo (left-aligned, clickable → returns to this home page)
- Help Button: Redirecting to FAQs and useful documentation

- *Menu Item: Containing different items (e.g. settings, Profile info, History, Log-out)*

b) Main Banner / Dashboard Area (Disputes Section)

A central, responsive content area containing user-specific data. The default view after login focuses on Dispute Management.

Key Elements:

- Disputes overview
- Editing
- Deleting

c) Footer

A lightweight, static bar anchored at the bottom of the screen.

Key Elements:

- Legal Links
 - Terms of Use
 - Privacy Policy
- Project Info
 - CREA3 project description or link to the external site
- Partner Logos (small size, grayscale)
- Contact Email or Support Link

Accessibility

- High-contrast text
- Fixed position for ease of reference

e) Chatbox Popup

Offer users contextual help, innovative suggestions, and continuous real-time support through an AI Assistant.

Design

- Appears as a circular chat icon in the lower-right corner
- On click: expands into a modal or side panel chat window

Behavior

- Chat history retained per session with time stamp and interlocutor name
- It's fixed chatbox accessible at any time while user is in the platform and can be minimized back to icon at any time.

3.1.5 Create New Dispute

Allow logged-in users to initiate new disputes; the fields should display a (?) sign explaining their relevance.

a) Access

- From Disputes Panel → "Create New Dispute"

Key Elements:

- Dispute Name
- Resolution Method
- Rating Weight
- Dispute Type:
 - Rating Method
 - Star weighting (1-5 stars)
 - Bidding Method
- Add assets and agents (optional for now or later)

b) Backend

Flow

- Saves the draft if incomplete.
- Validates resolution method compatibility with type.

c) Postcondition

- **Validate:** Redirect to Dispute Details with a confirmation message.
- **Save & Quit:** save the draft for future modifications.

3.1.6 Managing Existing Disputes

Enable users to edit or remove previously created disputes (if the dispute is not ongoing)

a) Access

- From Disputes Panel→ Delete
“Are you sure you want to delete this dispute? This will remove access for all agents involved.”
- From Disputes Panel→ Edit
Edit view pre-fills all current fields.

b) Functionality

- Change resolution method (if no evaluations have started)
- Change assets and agents (if no evaluations have started)
- Rename dispute
- Adjust weights or parameters

3.1.7 Agent Management

Enable the dispute creator to add, edit, or remove agents (in the Procedure Initialization phase). The agents SHOULD be regularly registered to the platform, and the dispute should not be in proceeding (once initialized, it is not possible to modify it or delete it).

a) Access

- From Disputes Panel→ Edit→ Agents Tab

b) UI

Behavior

- Table of current agents (name, email, quota)

- Add Agent modal (input email + optional quota %)
- Delete Agent via trash icon
- Tooltip: “If no quotas are defined, equal shares are applied.”

c) Validation

- Email must correspond to a registered user.
- Total quota must not exceed 100%.

3.1.8 Goods & Assets Management

Add and manage disputed assets.

a) Access

- From Disputes Panel → Edit → Asset Tab

b) Interface

- Table: Asset Name, Estimated Value, Divisibility (Y/N)
- Buttons:
 - “Add Asset”
 - “Edit”
 - “Delete”

c) Add

Specifics

- Name
- Value (numeric input with currency)
- Shareable / Invisible (checkbox)

d) Behaviors

Edits are allowed until the dispute enters evaluation and the Initialization procedure ends.

3.1.9 Validation Phase

a) Access

- Only available to moderators.
- The “Start Evaluation” button becomes visible once agents and assets are defined, with the initial agent presenting at least two counterparts and two assets to share.

b) Flow

Sends platform invitations to agents.

3.1.10 Video Conference Integration

Let users initiate videocall with other participants in the dispute.

a) UI/UX

- In Dispute Details: “Start Video Conference”
- Input: agent, mediators selection from list

b) Postcondition

- Opens an in-app video window or an external link (Zoom, Meet, or Jitsi integration).
- Warning shown:
“Please do not share private data during video calls.”

3.1.11 FAQs & Help

Clarify domain-specific terms and FAQs that help users throughout all phases of the disputes.

a) Access

Points

- Question mark icon next to each term or field.
- The dedicated information page is accessible via the upper banner on the personal homepage [Sec. 3.1.4].

4. Knowledge Requirements

This section describes various use case instances of the chatbot assistant retrieval tools that will be included in CREA3 project aiming to provide specific legal information on divorce and inheritance disputes (diverse user cases are provided to demonstrate the chatbot's expected functionality in various scenarios, highlighting its ability to provide the most relevant and accurate advice as presented in Fig.2) as well as static general assistance tools (FAQs, Handbooks, other documentations) assisting the user along each step of the dispute resolution directly on the platform.

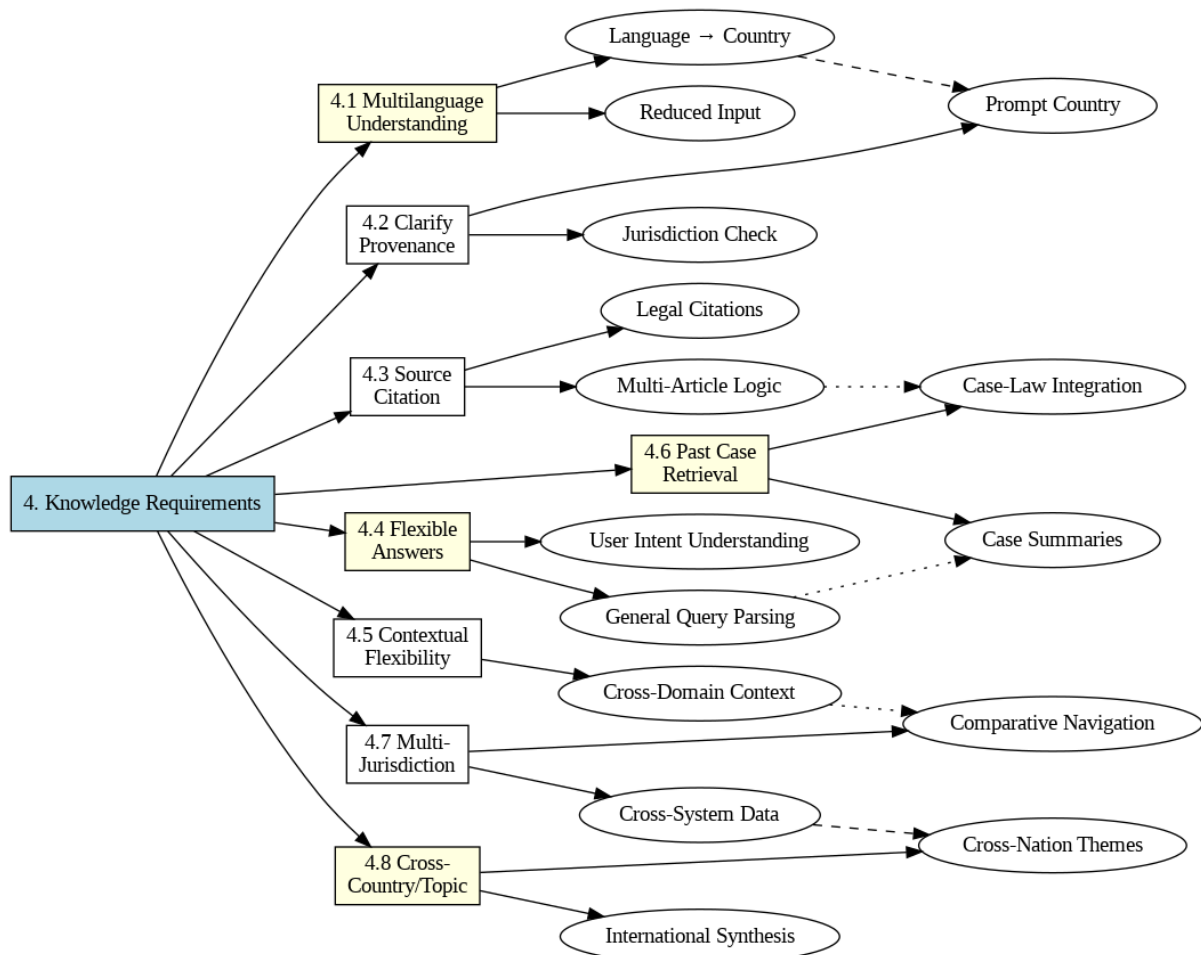


Fig.2 Schematic representation of CREA3 AI assistant requirements

4.1 Multilanguage Understanding

One of the key abilities of the CREA3 assistant is its ability to infer the relevant jurisdiction purely from the user's query. When a query is submitted in a language uniquely associated with a specific nation—the assistant automatically maps that language to the corresponding country and retrieves jurisdiction-specific legal content. In the CREA3 framework, where ease of use and cross-border interoperability are essential, this mechanism streamlines legal assistance for citizens and legal professionals operating within the consortium partner countries.

4.2 Clarifying User Provenance When Language Alone is Ambiguous

In cases where a language is used across multiple countries (e.g., English, French) or is ambiguous, the CREA3 assistant is designed to handle this uncertainty by prompting the user for clarification, ensuring that the legal guidance provided is both accurate and relevant and preventing misinterpretation while ensuring compliance with national legal frameworks. This step is essential in a system like CREA3, where multilingual inclusivity must be balanced with legal precision across member states.

Note: While the assistant can determine the country from language, it does not translate or generate multilingual responses unless preloaded with content in the target language. Response localization remains a design choice based on the pre-curated CREA3 knowledge base. CREA3 chatbot is not directly responsible for multilanguage respon

4.3 Source Citation Support

Transparency is a foundational principle in legal contexts, and the CREA3 chatbot assistant incorporates this by providing precise references to the legal documents and civil code articles used in its responses. Practically, when handling questions, the AI assistant SHOULD retrieve multiple relevant sources, citing them clearly and even summarizing their content to satisfy user queries. This feature builds trust and supports traceability, which is critical for users needing to verify or consult legal professionals afterward. Moreover, the AI system MAY manage more complex queries involving multi-legislation topics by identifying similar patterns from different parts of a codebase.

4.4 Answer Flexibility for Non-Specific Queries

A key strength of the CREA3 assistant is its ability to handle vague, open-ended, or exploratory questions. Users are not required to know specific legal terminology or reference numbers. Instead, the assistant parses intent and responds with relevant information by dynamically scanning its legal knowledge base. Whether the query is general ("How does divorce work in Europe?") or context-specific ("What happens if there is no will in my country?"), the assistant SHOULD offer accurate, filtered responses. This is particularly valuable according to CREA3's objectives.

4.5 Contextual Flexibility with Cross-Domain Argument Retrieval

Legal queries often span more than one domain—particularly in family law, where divorce issues intersect with inheritance, custody, or property rights. CREA3 addresses this complexity by equipping the assistant with the ability to retrieve and reconcile content across domains. The assistant SHOULD retrieve personal property division, child support, and succession rights content. Cross-argument handling reduces fragmentation and increases coherence, which is critical for supporting holistic legal navigation.

4.6 Cross-Document Retrieval with Past Case Integration

Going beyond statutory texts, the CREA3 assistant SHOULD be capable of integrating insights from real-life legal cases into its responses, summarizing past court rulings, highlighting procedural trends (e.g., cost ranges and average duration), and clarifying expected legal outcomes based on the user query. Doing so bridges the gap between codified law and applied practice.

4.7 Retrieval of Multiple Cases Across Themes and Jurisdictions

The CREA3 assistant is designed to operate at the intersection of multi-thematic and multi-jurisdictional legal information. It can retrieve relevant data across multiple legal themes, such as divorce, inheritance, and from several jurisdictions (related to consortium partners participating in the CREA3 project). This allows users to understand comparative legal approaches across countries—an essential feature in contexts such as cross-border family law, migration, or EU citizenship.

4.8 Cross-Country, Cross-Topic Document Integration

In its most advanced form, the CREA3 assistant SHOULD synthesize content from different countries and legal categories in a single interaction exemplifying the CREA3 initiative's core principles of legal interoperability, access to justice, and user empowerment.

4.9 FAQs and Handbook

This section presents the most frequently asked questions and the related answers, which form a static knowledge base for users regarding the following topics:

1. General information about CREA Project & User Account [[Sec. 4.9.1](#)]

2. Disputes Management & Details [\[Sec. 4.9.2\]](#)
3. Other tools (chatbots, video conferencing) [\[Sec. 4.9.3\]](#)

4.9.1 General information & User Account

General information and user account panels are needed to help with legal or technical jargon, giving an immediate grasp of what CREA3 is and how to get started with the dispute initialization. This information can be grouped into three main categories:

- **Platform overview**
A one-paragraph “CREA3 in a nutshell” that explains the AI-driven dispute-resolution flow and its EU-wide scope.
- **Using the system**
The step-by-step case workflow: inviting parties, listing goods, stating preferences, and receiving the algorithm’s proposed allocation.
- **Account Lifecycle**
Practical how-tos for signing up, accepting an invitation, editing personal data or password, and deleting an account (with the caveat that deletion isn’t possible during an active dispute).

Topics	Summarized content
Platform overview	CREA3 in a nutshell – An EU-focused online platform that applies AI, game theory, and blockchain to guide people and businesses through civil-dispute resolution, ending with an algorithm-generated fair split of the disputed goods.
System Usage	How a case works – • Invite the parties • List each good with its value • Express your preference on every good • The system proposes an equitable allocation
Account basics	Sign-up – Create an account from the login page to access CREA3 Join a dispute – Follow the e-mail link, log in (or register), and accept the invitation. Edit details – Update name or password anytime in profile settings. Delete account – Possible from settings, except during an active dispute; deletion is permanent.
FAQs indication	1. What is CREA3? 2. How do I create an account on CREA3? 3. I’ve received an e-mail to join a dispute—how do I accept the invitation? 4. Can I edit my account information on CREA3? 5. Can I delete my account on CREA3?

4.9.2 Disputes Management

This category of FAQs helps users navigate the key elements of the process, from understanding what a dispute entails to finalizing a certified agreement. The information is organized into clear categories to guide both new and more expert users through each stage of the dispute workflow.

Topics	Summarized Content
What is a dispute?	A dispute on CREA3 is a structured legal procedure to resolve conflicts over asset division. The platform guides users with AI tools and game-theoretical algorithms to help all parties reach a fair, efficient, and tailored resolution.
Resolution methods	What are the resolution methods "Bids" and "Rates"? • Bids Method – Users distribute virtual money across goods to reflect their preferences. • Rates Method – Users rate each good on a 1–5 star scale to show importance.
Inviting agents	How do I invite agents to join a dispute? • Go to the agent section of your dispute setup. • Click "Add New Agent". • Fill in their name, email, share of entitlement, and optional role. • Submit to add them.
Adding goods	How can I add goods to a dispute? • During setup, navigate to the goods section. • Click "Add New Good". • Enter the good's name, and estimated value, and optionally mark it as indivisible. • Submit to include the good in the dispute.
Agent preferences	Agent preferences reflect how each agent values the goods in the dispute. These are input using either the Bids or Rates method selected during dispute setup.
Resolution process	How does CREA3 provide a solution to my dispute? CREA3 uses agents' preferences and applies AI and game-theoretical algorithms to generate an allocation that aims to fairly satisfy all parties.
Finalizing a dispute	What happens once all agents agree on the proposed solution? When all parties accept the solution, CREA3 finalizes the case. A downloadable report is generated as certified proof of resolution, backed by Smart-contract Blockchain technology.

4.9.3 External Tools

In addition to its core dispute resolution functionalities, the CREA3 platform provides a suite of supplementary tools designed to enhance user experience, streamline the negotiation process, and ensure transparent, efficient conflict resolution.

Mainly, the issues that users can observe are related to the video conference and AI assistant.

Topics	Description
Videoconference tool	What is the videoconference feature on CREA3? • Enables real-time meetings between agents, users, and optionally mediators. • To create a meeting: click "videoconference", set a date/time, optionally add a mediator and agents, then click "Send invitation."
Invitation limitations	I can't invite an agent to the video conference. Is this normal? • Yes. You can only invite agents who have joined the dispute and completed the first step. Once validated, they appear in the invitation list.
AI chatbot assistance	How can the chatbot help me during a dispute? • Legal Guidance – Offers explanations about rights and obligations. • Procedure Clarification – Helps with next steps. • Document Assistance – Supports document prep. • Regulation Insights – Informs about laws. • Terminology – Explains legal jargon.

6. Performance requirements

Almost immediate response times are required; the AI assistant needs to give the user the impression that they are talking to a real person. So, a minimum time of a couple of seconds should be waited to type a response.

6.1 Multi-user Asynchronous

The AI assistant **MUST** handle conversations with thousands of users simultaneously and offer enough power to control them all without slowdowns or disruption. All procedural metadata, user profiles, and conversational agent logs **SHOULD** be stored in a relational database (or other types) that enforces ACID properties and supports multi-language collation. Evidentiary files shall be referenced through content-addressable hashes and preserved in immutable object storage. Personal data **MAY** be segregated from public case metadata through logical separation and field-level encryption. The schema **MAY** expose audited APIs suitable for integration with national e-justice portals.

6.2 Security

The chatbot software system for CREA3 processes confidential information that SHOULD be accessible only to mediators and agents who upload it. Each user can view their personal information by accessing the platform using the username and password provided during registration for the CREA3 platform.

6.3 Privacy & Fairness

One of the main aims of the CREA3 advancements and scope is to address fairness criticalities, promoting the safe and trusted usage of the AI-driven conflict resolution platform. The fundamental features of this trustworthy application, such as privacy issues, source selection and algorithm explainability, data bias, and model fairness, carry significant weight and will be carefully considered throughout the development phase. From a data security perspective, which is devoted to privacy issues concerning any information or reference to the subjects of past legal cases in the algorithm knowledge base, the data MUST be previously processed and coherently anonymized under the directives of the General Data Protection Regulation (GDPR), ensuring full compliance and rendering impossible any data leakage.

The security-by-design approach will be complemented by the proactive nature of Prompt Engineering, filling possible anonymization gaps and guiding the LLMs to avoid focusing on subjects' information if found.

The trustworthiness of the system SHOULD be granted by the source selection and the algorithm's explainability, which involves inspecting its "reasoning" and visualizing the source metadata attached to all content in the knowledge base.

6.4 Accessibility

Special attention is given to assessing the needs of specific user groups, such as the visually impaired or those with dyslexia, to ensure inclusivity and accessibility within the system. The adopted criteria for non-discriminatory accessibility are organized to create a more accessible web for everyone, including those with disabilities. The CREA3 accessibility pillars are directly based on the following four WCAG principles, which are referenced in works by Reid et al. [4] and the WCAG 2.1 standards [5].

The CREA3 platform SHOULD be:

- Perceivable - Information and user interface components must be presentable to users in ways they can perceive.
This means that users must be able to perceive the information being presented (it can't be invisible to all of their senses)[5]
- Operable - User interface components and navigation must be operable.
This means that users must be able to operate the interface (the interface cannot require interaction that a user cannot perform)[5]
- Understandable - Information and the user interface must be understandable.
This means that users must be able to understand the information as well as the operation of the user interface (the content or operation cannot be beyond their understanding)
- Robust - Content must be robust enough that it can be interpreted reliably by a wide variety of user agents, including assistive technologies.
This means that users must be able to access the content as technologies advance (as technologies and user agents evolve, the content should remain accessible)

More information about the Crea3 platform accessibility standards:

- Goal:
The CREA3 platform has to cover a broad range of disabilities, including visual and cognitive.

- Versions:
CREA3 MAY adopt WCAG 2.1 as a development standard.
- Conformance Levels:
WCAG defines three levels of conformance: A, AA, and AAA.
The CREA3 platform MAY satisfy the AA level, which is generally considered the minimum standard for most applications.

7. Use-Case Catalogue

A detailed list of use cases, each mapped to a primary actor, precondition, main flow, and postcondition, will follow in this section of the document, presenting a series of critical conditions for information retrieval via chatbot assistant [3].

Case Study I: Multilanguage Understanding a)

Aspect	Details
Scenario	The system receives a user query written in a specific language without an explicit mention of the country.
User Goal	To obtain legal information relevant to their country without needing to specify it directly. (User / Legal Jargon, mother language speaker)

System Role	The system automatically detects the language of the user's query (e.g., Estonian, Slovenian) and infers the appropriate country to provide relevant legal guidance.
Outcome	Users receive accurate and localized legal information seamlessly, improving accessibility and reducing the need for manual input. the response MAY be in the same language of the user query or in the same language of the legislation required (this feature is not guaranteed by the CREA3 AI assistant)

Case Study II: Multilanguage Understanding b)

Aspect	Details
Scenario	A user submits a legal query in French, a language spoken in several countries, without specifying a location.
User Goal	To receive accurate legal information about relevant divorce. (User / Legal Jargon)
System Role	The system detects the ambiguity and prompts the user to clarify the country (e.g., "Belgium") to ensure accurate jurisdictional information is provided.
Outcome	Once the user confirms the country, the system delivers precise legal guidance for that jurisdiction, demonstrating its ability to handle multilingual ambiguity effectively.

Case Study III: Chatbot Support for Legislative Guidance

Aspect	Case 1: Inheritance Between Siblings (More Specific)	Case 2: Inheritance Contestation on Father's Estate (More General)
Scenario	The user asks how inheritance is handled between siblings when no will is present.	The user seeks guidance on how inheritance disputes are managed when there is no will from the father.
User Goal	To understand how property is divided among siblings in different configurations. (User / Legal Jargon)	To understand what legal processes apply when contesting the division of a deceased parent's estate. (User / Legal Jargon)
System Action	The assistant identifies the context and provides coherent references (specific articles) relevant to the required context.	The assistant provides a broader legal overview (including more articles) relevant to disputes and contested inheritances without a will.

Outcome	The user receives clear and specific guidance on inheritance distribution.	The user gains clarity on the steps and procedures involved in handling broader inheritance contests.
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Case Study IV: Flexible Legal Query Handling for Multiple European Countries

Aspect	Details
Scenario	The user requests general information about how divorce is regulated across various European countries.
User Goal	To explore legal frameworks for divorce in different European contexts, aiming to understand how property and marital obligations are handled. (Legal Practitioner)
System Action	The assistant first requests clarification but proceeds to provide a comparative overview of divorce regulations in multiple countries (e.g., Belgium, Croatia, Estonia, Italy, Lithuania, Slovenia). Upon further request, it delivers specific legal references from each country mentioned.
Information Scope	The assistant draws exclusively from the CREA3 knowledge base and ensures the responses remain legally relevant and jurisdiction-specific, without referencing external or unrelated jurisdictions.
Outcome	The user receives a clear, organized comparison of divorce laws and relevant legal articles across selected European countries, demonstrating the assistant's ability to handle general, open-ended queries effectively and contextually.

Case Study V: Cross-Context-Country Information Retrieval

Aspect	Details
Scenario	The user, identifying as being from Italy, asks for civil code articles related to both inheritance and divorce regulation.
User Goal	To obtain official references from the Italian Civil Code that specifically address inheritance and divorce matters. (Legal Practitioner)
System Action	The assistant searches across multiple legal topics and retrieves a set of relevant articles from different sections of the Italian Civil Code, effectively bridging multiple legal domains (inheritance and divorce).

Information Scope	The assistant provides a curated list of articles along with readable summaries, presenting information on both transactional inheritance rules and the administration of personal and community property in divorce.
Outcome	The user gains access to accurate, multi-topic legal information in one query, demonstrating the assistant's ability to retrieve and summarize cross-domain legal content efficiently.

Case Study VI: Cross-Document Legal Retrieval – Past Legal Cases

Aspect	Details
Scenario	The user from Lithuania requests legal information on divorce, including references to relevant civil code articles and insights from past legal cases (e.g., costs, case duration, and issues addressed).
User Goal	To understand not only the legal framework but also the practical and procedural aspects of divorce cases in Lithuania. (User / Legal Jargon)
System Action	The assistant retrieves statutory content from multiple articles of the Lithuanian Civil Code and supplements this with a summary of a real past case. It details financial implications, procedural steps, and the type of issues addressed during litigation.
Information Scope	The response combines civil code statutes with contextualized knowledge from actual divorce case documentation, offering a comprehensive legal and procedural overview.
Outcome	The user gains valuable insights into both the legal requirements and practical realities of divorce, including typical costs, obligations, and court expectations, demonstrating the system's advanced cross-source integration capabilities.

Case Study VII: Cross-Context-Documents-Country

Aspect	Details
Scenario	The user requests legal information on both divorce and inheritance laws across European countries of the assistant's choosing, along with relevant articles and past case examples.
User Goal	To understand how divorce and inheritance matters are legally addressed in different European jurisdictions, supported by official articles and real-case context. (Legal Practitioner)

System Action	The assistant autonomously selects Belgium and Italy, retrieves related articles from their respective legal codes, and provides illustrative case summaries, including duration and key legal aspects.
Information Scope	The system presents articles on divorce and inheritance from both countries, referencing joint property disputes, hereditary estate divisions, and judicial procedures—along with real-life case timelines.
Outcome	The user receives a cross-jurisdictional overview that merges civil code references with contextualized insights from past cases, reflecting the assistant's capacity for non-targeted, multi-theme legal knowledge synthesis.

References

- [1] IEEE Recommended Practice for Software Requirements Specifications IEEE Std 830-1993 (Revision of IEEE Std 830-1984).
- [2] Key words for use in RFCs to Indicate Requirement Levels. (RFC 2119).
- [3] Cirillo, E., Giacalone, M., Moccardi, A., Pelosi, M. (2025). Ethical and Technical Challenges in the European Conflict Resolution Project: The Role of Equitative Algorithm (CREA3). In: Barolli, L. (eds) Advances in Internet, Data and Web Technologies. EIDWT 2025. Lecture Notes on Data Engineering and Communications Technologies, vol 243. Springer, Cham. https://doi.org/10.1007/978-3-031-86149-9_40
- [4] [Web Content Accessibility Guidelines \(WCAG\) 2.0](#). Ben Caldwell; Michael Cooper; Loretta Guarino Reid; Gregg Vanderheiden et al. W3C. 11 December 2008. W3C Recommendation. URL: <https://www.w3.org/TR/WCAG20/>
- [5] WCAG standards: <https://www.w3.org/TR/2024/REC-WCAG21-20241212/>